Şevket Kaan Alkır

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About Me!

Master's candidate in Mathematics at Bilkent University with a B.Sc. from Ankara University. My research spans inverse reinforcement learning and mean-field games under an average-reward criterion; I've submitted a journal paper to IEEE CDC 2025 and am currently developing kernel-based methods. As a Math Department TA, I've supported courses every semester.

Education

Bilkent University, *M.Sc. in Maths* — GPA: 3.01/4.0 — *2023–Present*

- *In my thesis, I study inverse reinforcement learning for mean-field games under an average-reward criterion. I employ a maximum causal entropy approach to recover expert-consistent policies. Also, model the reward function in a reproducing kernel Hilbert space to capture flexible functional structures.*

Ankara University, *B.Sc. in Maths* — GPA: 3.44/4.0 — *2018–2023*

Research Interests

- Reinforcement Learning.
- Inverse Reinforcement Learning.
- Stochastic Optimization and Control.
- Game Theory in Multi-Agent Systems.
- Applications of RL in Complex Systems.
- Decision Theory under Uncertainty.

Experience

Trainee — *Prague, Czech Republic*
Charles University — *June – Sept. 2022*

- Used resources "Lectures on Choquet's Theorem, Robert R. Phelps" and "Compact Convex Sets and Boundary Integrals, Erik M. Alfsen"

Teaching Assistant — *Ankara, Turkey*
Bilkent University — *Oct. 2023 – Present*

Publications and Manuscripts

Inverse Reinforcement Learning for Mean-field Games with Average Reward Criterion — *Dec. 2025*
*Şevket Kaan Alkır**, Naci Saldı. *IEEE Conference on Decision and Control*

Kernel Based Inverse Reinforcement Learning for Mean-field Games with Average Reward — *in preparation*
*Şevket Kaan Alkır**, Naci Saldı. *Manuscript in Preparation*

Presentations and Talks

Full list of talks with abstracts and materials

Inverse Reinforcement Learning for Mean-field Games — *April 2025*
Bilkent University

Average-Cost Mean-Field Games: Forward and Inverse Perspectives — *June 2025*
Özyeğin University

Grover's Search Algorithm — *Dec. 2024*
Bilkent University

Frobenius Algebras — *Dec. 2023*
Bilkent University / Representation Theory Class

Multi-Criteria Decision Making: SMART and VIKOR Methods — *June 2022*
Ankara University

Awards and Scholarships

Department of Mathematics Graduate Scholarship *Sep. 2023 - Present*
Bilkent University

1001–Scientific and Technological Research Projects Support *Oct. 2024 - Present*
Scientific and Technological Research Council of Turkey (TUBITAK)

Projects

Kalman Filter State Estimation *GitHub*

- Simulated Kalman filter convergence in Python using Riccati updates and visualized estimation errors.

2D Incompressible Fluid Simulation *Simulation*

- Implemented a numerical solver for the incompressible Navier–Stokes equations using semi-Lagrangian advection and pressure projection on a uniform grid.

Multi-Criteria Decision Making *Document*

Workshops and Events

Probability and Applied Analysis Summer School *June 30 - July 12, 2025*
Boğaziçi University

Stochastic Days 2025 *June 2025*
Özyeğin University

Stochastic Days 2024 *June 2024*
Middle-East Technical University

Academic References

Naci Saldı, *Bilkent University* *naci.saldi@bilkent.edu.tr*
Serdar Yüksel, *Queen's University* *yuksel@queensu.ca*

Skills

Languages: English (IELTS Band 7), Turkish (Native).
Programming Languages: Python, MatLab, JavaScript. [1] [2]